Morphing Airfoil

Warping Wing Concept

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Senior Design Report — MAE 4291

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1 Executive Summary

1.1 Overview

This report documents the design and development of a morphing trailing-edge mechanism for RC-scale fixed-wing aircraft. The concept replaces a conventional hinged control surface with a continuously deforming rib structure that shifts wing camber while keeping all actuation hardware enclosed within the airfoil profile. The following questions outline the design process, analytical methods, and prototype; the report then continues to a more detailed account of each.

1.2 MAE Department Questions

1. **What are the desired function(s) of your design?**

   Two primary functions drove this design. First, develop a fully enclosed mechanism for changing wing camber. Second, replace discrete control surfaces with a continuous morphing alternative.

2. **What constraints related to the main function(s) must your design satisfy?**

   The mechanism must deflect to target angles and hold position under aerodynamic and mechanical load. All actuation and attachment hardware must stay within the airfoil profile. The rib design must tile repeatably along the wingspan to fill a target span.

3. **What are the performance objectives of your design?**

   Maximum downward and upwards deflection of **30 deg**. Full 60-degree range traversed in under **1 sec**. Hinge deflection spread over at least **10 mm**.

4. **What alternative design concepts were considered?**

   All serious candidates were compliant mechanism designs where part of the upper airfoil surface deflects to shift camber, with the lower surface accommodating the resulting gap. The main design exploration focused on two variables: skin material and actuation placement. PLA, PETG, TPU, and natural rubber were all considered as candidates for the bending surface. Servo placement and push rod attachment varied across a range of chord positions and airfoil thicknesses.

5. **What analyses were used to select among these alternative design concepts?**

   Three methods drove the down-selection. Material fatigue was assessed through repeated deflection cycles, tracking where creases and defects formed. Bending resistance was evaluated qualitatively by feel during manual deflections, serving as a rough but fast discriminator between candidates. Aerodynamic performance was evaluated in XFOIL, sweeping lift and drag coefficients across deflection angles.

6. **What industry or society standards were used to inform or evaluate your design?**

   Radio-controlled drones do not typically have industrial or societal standards to meet in their design, and are mainly constrained by flight-zone regulations. However, several rules of thumb common among model-aircraft pilots guided the design. The most significant was the “30-30” rule, where a maneuverable aircraft should have **30 deg** of deflection on control surfaces that make up **30%** of the chord length.

7. **Which concepts or skills learned in your coursework were applied to the design?**

   - **MAE 2020** — Several iterations of the design featured a compliant mechanism undergoing moment bending along the top side of the airfoil. The equations governing the material’s resistive torque were initially derived from course lecture notes.
   - **MAE 3050** — Understanding the application of airfoils in the broader context of wings and aircraft design helped direct the desired functions of the project and their potential implementations in existing small-scale aircraft.
   - **MAE 3230** — Modeling basic flow over an airfoil requires knowledge of potential flow theory. Concepts from this course were used to build up the aerodynamic analysis, inform hinge moment calculations for deflected airfoils, and provide context for simulations conducted in XFOIL.

8. **Evaluate your design relative to its function(s) and constraints.**

   The first objective, moving all actuation fully inside the airfoil, was met. The entire mechanism fits within the profile, with one continuous outer covering across the wing. The second objective, continuous smooth morphing, was only partly achieved. The bottom surface gap is adequately covered across the deflection range, but the top surface required a physical hinge point, which produces a kink rather than smooth curvature. Compared to published morphing-wing implementations, the gap-coverage approach is consistent with common practice, but those designs typically maintain a kink-free top surface, likely through different rib materials or outer film stiffnesses.

9. **What impact do you see your design having upon public health, safety, and human welfare?**

   An effective morphing-wing design reduces the drag losses associated with traditional hinge gaps and enables beneficial camber control across the wingspan, improving performance near stall and in cruise. Fixed-wing drones are currently deployed in environments ranging from firefighting to agriculture, where aerodynamic efficiency directly affects endurance, range, and maneuverability. As fixed-wing aerodynamics improve, so does their utility in critical applications.

10. **What format did your design take?**

    The final design is a **240 mm × 300 mm** working prototype of a morphing-wing segment with a rigid rib structure, smooth outer wrapping, and an expanding outer perimeter to accommodate camber changes. Excluding either end of the segment, the camber-shifting mechanism is fully enclosed within the airfoil perimeter throughout the full range of motion.

11. **Describe each student's role in the design project.**

    This was an individual project, though multiple members of the DBF project team provided feedback and opinions throughout the design process.

2 Motivation

Wing warping was a staple of early aircraft, featuring a system of pulleys and cables to twist wingstips in opposite directions. The gradual twist allowed pilots to roll their aircraft without requiring discrete flow disruptions on wing surfaces. Eventually, wings became stiffer and incorporated ailerons, with their simplicity outweighing the aerodynamic downsides.

Modern drones, however, present a unique opportunity to reanalyze these control surfaces. Being built consistently smaller and more power efficient, incorporating newer materials with novel designs stands to achieve the benefits of wing warping without compromising manufacturability.

Wing warping can be achieved by successively shifting the camber of each wing rib across a drone's wingspan. Having precise control over this effect offers multiple benefits beyond the baseline drag reduction of a continuous wing surface. Decreasing the wing incidence angle away from the fuselage - often referred to as washout - delays stall on the wingtip, promoting controllability in high-load maneuvers. In other conditions such as cruising, camber can be shifted across the wing to follow an efficient elliptical lift distribution. At higher speeds, rolling an aircraft can also create adverse yaw, where the higher-lift wing experiences more drag than the other. Tailoring and spreading out the lift necessary for rolling decreases the drag penalty in roll, decreasing the adverse yaw.

3 Design Requirements

Although wing warping presents many benefits, ailerons are still predominant on aircraft and fixed-wing drones. To guide the design process, the following design requirements were established in **Table 1**.

Requirement	**Value**	**Justification**
Downwards deflection	**30 deg**	Maximum desired deflection for flaps in landing configuration
Upwards deflection	**30 deg**	Standard deflection in maneuverable RC aircraft, eletronically adjusted if necessary
Travel time	**1 sec**	Typical greatest input lag from full deflections on human-piloted fixed-wing drones
Constraint		
Self-contained actuation		Opening the wing to external flow defeats aerodynamic gains
Continuous profile		Removing kinks in wing surface promotes attached airflow down the trailing edge
Design tiling		Design should be easily scalable and repeatable along the wingspan

Table 1: Requirements and constraints

4 Preliminary Analysis

4.1 Airfoil Selection

Airfoil selection started with one constraint: the baseline profile needed positive camber, so the wing generates lift in cruise without a constant downward bias on the morphing surface. A symmetric airfoil would shift that bias onto the actuator, eating into the deflection budget in one direction. From the airfoils already used by the DBF team, the USA-35B was the practical pick; its flat lower surface simplified the gap-sealing geometry and gave clean pushrod attachment points along the bottom skin.

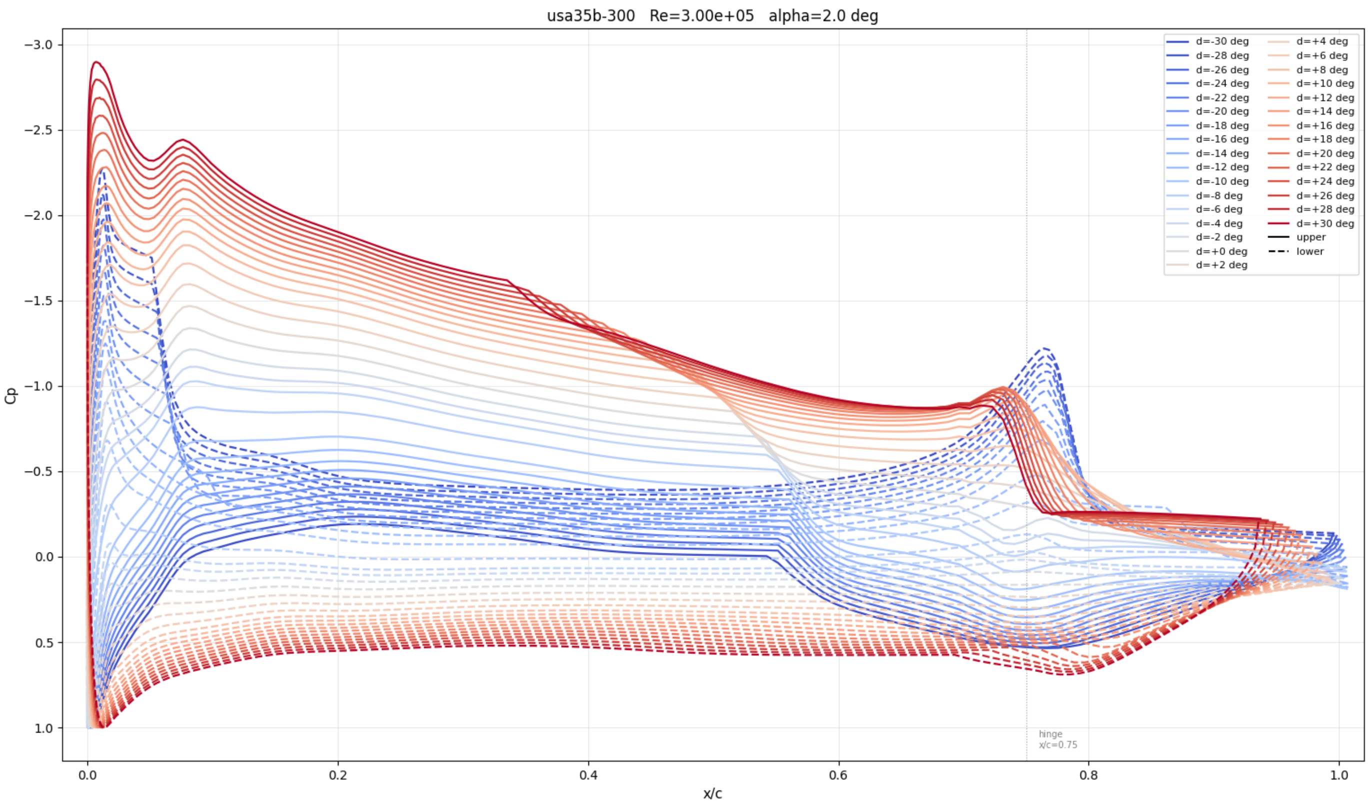

Figure 1: Pressure distribution sweep at variable deflection angles

4.2 Torque Modeling

To set the torque requirements early, material and aerodynamic models were built to predict the loads a sustained deflection would produce at cruise speeds. The approach started with USA-35B, placed a hinge on the upper surface at the desired chord location, then reworked the spline to smooth the resulting geometry. That smoothed profile was imported into XFOIL, which ran a vortex panel method at the expected Reynolds number to extract the surface pressure distribution shown alongside the deflected geometry in **Figure 1**. From that pressure distribution, combined with the rib's geometric and material properties, the reactive bending moment could be calculated and summed into the total restoring moment, given in **Equation 1**, where $\Delta C_p = C_{p,\text{lower}} - C_{p,\text{upper}}$, x_h is the hinge location, b is rib thickness, and c is chord length.

$$M_{total} = \frac{EI}{L_{flex}} \cdot \delta + q_\infty \cdot b \cdot c^2 \int_{x_h/c}^{1} \Delta C_p(\xi)\,(\xi - x_h/c)\,d\xi \tag{1}$$

5 Design Concept

The initial concept centered on a compliant trailing section that deflects up when pushed and down when pulled. As shown in **Figure 2**, the forward portion of the rib and the trailing edge would be built as truss structures to concentrate deformation within the morphing region, sized at roughly **15%** of chord. The servo sits in the thickest part of the airfoil with its arm oriented upward to follow the upper surface curvature and connects to the trailing edge via an aluminum pushrod pinned through a small hole in the aft section.

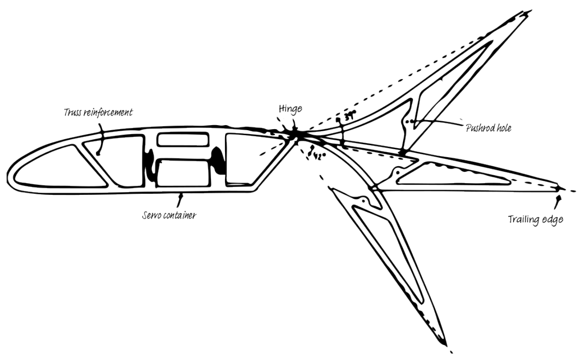

Figure 2: Initial design sketch

Three problems were apparent from this concept. The geometry left a large gap on the lower surface, resulting in more elastomeric skin required to cover it. The morphing region and deflection targets were also oversized relative to what the servo budget could realistically support. Finally, because the pushrod runs out of the rib plane, any compliance in the linkage would produce an out-of-plane component in the trailing edge motion alongside the intended vertical deflection. These issues defined the targets for the following iterations.

6 Down-Selection

Each iteration was driven by three factors: reducing the lower surface gap, keeping the total restoring moment within the servo's torque budget, and eliminating out-of-plane motion in the linkage. **Table 2** summarizes the key design changes across iterations and the insight each one produced.

Iteration	Investigation	Conclusion
4	Top surface thickness	A **1.5 mm** surface thickness in standard PLA firmly holds the trailing edge while not creating too much resistance in deflection.
5	Control surface chord percentage	A morphing region spanning **20–30%** chord from the trailing edge achieves the target deflection range while keeping the compliant section compact.
6	Pushrod pin position	Relocating the pushrod pin to the lower surface increases the moment arm, improving both available torque and deflection range.
9	Rib thickness	An **8 mm** rib with a **4 mm** pushrod holder constrains the trailing edge against out-of-plane deflection.
11	Rubber hinge feasibility	A **12 mm** transition region balances deflection smoothness against trailing edge retention.
12	Bottom surface material	A pre-tensioned latex covers the bottom surface gap, but requires a high-torque servo to sustain a full range of motion.
13	Segment prototype	Comb cutouts enable repeatable span-wise assembly; CA-bonded hinges stiffen the trailing edge connection.

Table 2: Key iterations and design conclusions

The final rib integrates three angled comb features into the truss structure (one at the leading edge and two mid-chord) for repeatable span-wise assembly. The servo sits horizontally in the thickest part of the airfoil, with its arm reaching the top of the profile without restricting the deflection range. A pushrod connects the arm to a pin at the lowest point of the trailing edge section, maximizing the moment arm about the hinge.

The hinge itself required its own selection process. A fully compliant PLA flex zone was the first candidate, but fatigue testing ruled it out quickly; PLA creases under repeated deflection, concentrating the bend at one point rather than distributing it smoothly. TPU would likely fix this, but material lead times made direct testing impractical. A rubber transition region depicted in **Figure 3** distributed the curvature well, but became too compliant at the **12 mm** length needed to reach the trailing edge. A hardened paper hinge bonded at the interface between the two rigid halves resolved this: stiff enough to resist lateral play, flexible enough to allow the full deflection range without fatigue failure.

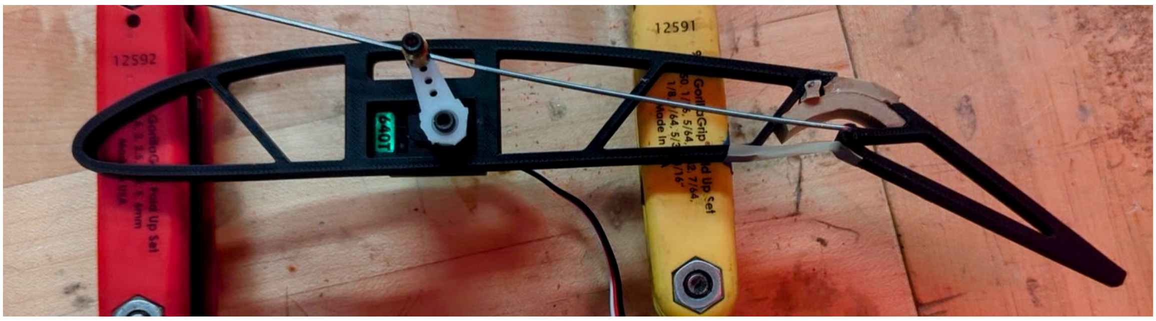

Figure 3: Iteration 12, rubber transition morphing airfoil variant

7 Prototype Evaluation

The prototype, shown in **Figures 4** and **5**, used the full **240 mm** chord and a **300 mm** span with four ribs spaced **100 mm** apart. The comb system from iteration 13 achieved consistent geometry across all three ribs, confirming that the design tiles repeatably along the span.

The design ultimately met all five design objectives. Measured deflections reached **30 deg** downward and **30 deg** upward, the full 60-degree range traversed in under **1 sec**, and the actuation mechanism remained entirely within the airfoil profile

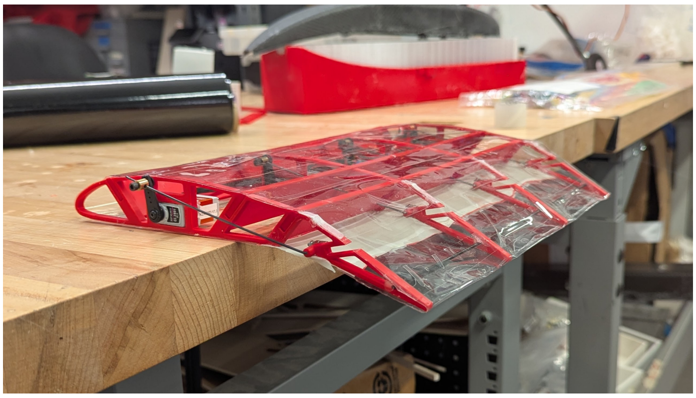

Figure 4: Final prototype side view

Figure 5: Final prototype bottom view

throughout. A single continuous skin covered the outer surface across the complete deflection range with no exposed gaps. A carbon fiber rod added to the trailing edge resolved the out-of-plane deflection problem that persisted through earlier iterations, constraining the trailing edge laterally across the full span without adding meaningful stiffness in the deflection direction.

Fatigue performance was evaluated by running the servo through the full deflection range continuously for 30 minutes via an Arduino-controlled actuation loop. No cracks, creases, or joint failures occurred. The latex skin on the lower surface showed visible stress concentrations at the attachment points but remained essentially smooth across the bay; there was no major rippling or delamination. The skin loosened slightly after the first few cycles as the latex settled, reducing resistance noticeably, but retained enough tension to stay taut and aerodynamically clean at full downward deflection.

The primary limitation was a visible kink at the paper hinge on the upper surface. Rather than distributing curvature smoothly across the transition region, the deflection concentrated at the hinge interface, producing a geometric discontinuity in the top surface profile. This likely increases drag relative to what the XFOIL analysis predicted for the smoothed geometry, though quantifying that penalty would require wind tunnel validation beyond the scope of this project.

Two observations point toward refinement in a next iteration. The three-comb truss system proved more redundant than necessary. The structure remained rigid under loading, suggesting the middle combs could be reduced without compromising stiffness. More significantly, the rib structure would benefit from transitioning to basswood in a production iteration. PLA was used throughout development for convenience since the Cornell RPL turnaround time was impractical for a rapid iteration cycle, but basswood would reduce weight, simplify tolerancing, and align the design with standard DBF construction practice. The core mechanism is unchanged by this substitution since no printed plastic component contributes to the distributed deflection behavior.

8 Conclusion

The prevailing assumption going into this project was that a fully enclosed morphing mechanism at RC scale would be too complex to be practical. The final prototype challenges that - the mechanism fits within the airfoil envelope, meets its deflection and response targets, and survives repeated actuation without structural failure. The feasibility has been largely confirmed; what remains is a materials problem, not a design one.

The binding limitation is the latex skin and hinge material stiffness. A less compliant elastomer at the hinge transition would reduce the restoring moment enough to bring the servo count down to a level practical for a DBF airframe. That material exists, but hasn't been identified and tested within this project's timeline. TPU for the flex zone and a lower-durometer rubber for the skin are the most immediate candidates. These two material questions separate the current prototype from a more flight-ready implementation.

The aerodynamic case for morphing over a conventional hinged flap is real but also unquantified; wind tunnel validation was outside the project scope. What the prototype does establish is that the integration problem is solved. A morphing trailing edge that fits within a standard RC airfoil profile, requires no external hardware, and tiles repeatably along the span is a viable starting point for a anyone willing to close the materials gap.